

CivicPulse

Version: 1.0 | Database Engine: PostgreSQL 17+ | ORM: Prisma ORM | Architecture: Relational Database

SUMMARY

DATABASE DESIGN DOCUMENT (DDD)

1. DATABASE DESIGN PRINCIPLES

Tujuan desain database: - Maintainable - Scalable - Auditable - Secure - Analytics Friendly

Prinsip: - UUID Primary Keys - Soft Delete - Audit Trail - Referential Integrity - Optimized Indexing - Future Multi-Tenant Support

2. CORE DOMAIN ENTITIES

Authentication Domain - User - Session

Organization Domain - Agency

Reporting Domain - Report - ReportAttachment - ReportComment - ReportHistory

Communication Domain - Notification

Monitoring Domain - AuditLog

Location Domain - Location

3. ENTITY RELATIONSHIP DIAGRAM (ERD)

Agency |— Users |— Reports

Report |— Attachments |— Comments |— History |— Notifications

User |— Reports |— Comments |— Notifications |— AuditLogs

Location |— Reports

4. USER ENTITY

Purpose Menyimpan seluruh akun pengguna.

Fields id UUID PK email VARCHAR UNIQUE password_hash VARCHAR full_name VARCHAR phone VARCHAR role ENUM agency_id UUID FK NULLABLE is_active BOOLEAN email_verified BOOLEAN created_at TIMESTAMP updated_at TIMESTAMP deleted_at TIMESTAMP NULL

Indexes email role agency_id

5. AGENCY ENTITY

Purpose Instansi pemerintah.

Examples Dinas PU Dinas Kesehatan Dinas Pendidikan

Fields id UUID PK name VARCHAR description TEXT email VARCHAR phone VARCHAR is_active BOOLEAN created_at TIMESTAMP updated_at TIMESTAMP

Indexes name

6. LOCATION ENTITY

Purpose Lokasi laporan.

Fields id UUID PK latitude DECIMAL longitude DECIMAL address TEXT district VARCHAR city VARCHAR province VARCHAR postal_code VARCHAR created_at TIMESTAMP

Indexes city district latitude longitude

7. REPORT ENTITY

Purpose Data inti laporan masyarakat.

Fields id UUID PK report_number VARCHAR UNIQUE title VARCHAR description TEXT category ENUM priority ENUM status ENUM citizen_id UUID FK agency_id UUID FK location_id UUID FK assigned_officer_id UUID FK NULL resolved_at TIMESTAMP NULL closed_at TIMESTAMP NULL created_at TIMESTAMP updated_at TIMESTAMP deleted_at TIMESTAMP NULL

Indexes status priority agency_id citizen_id category created_at

8. REPORTATTACHMENT ENTITY

Purpose Foto bukti laporan.

Fields id UUID PK report_id UUID FK file_name VARCHAR file_url VARCHAR file_size BIGINT mime_type VARCHAR uploaded_by UUID FK created_at TIMESTAMP

Indexes report_id

9. REPORTCOMMENT ENTITY

Purpose Diskusi antara citizen dan officer.

Fields id UUID PK report_id UUID FK user_id UUID FK message TEXT created_at TIMESTAMP updated_at TIMESTAMP

Indexes report_id user_id

10. REPORTHISTORY ENTITY

Purpose Audit perubahan status laporan.

Fields id UUID PK report_id UUID FK actor_id UUID FK old_status ENUM new_status ENUM note TEXT created_at TIMESTAMP

Indexes report_id actor_id created_at

11. NOTIFICATION ENTITY

Purpose Notifikasi sistem.

Fields id UUID PK user_id UUID FK title VARCHAR message TEXT type ENUM is_read BOOLEAN created_at TIMESTAMP

Indexes user_id is_read

12. AUDITLOG ENTITY

Purpose Audit seluruh aktivitas sistem.

Fields id UUID PK actor_id UUID FK action VARCHAR resource_type VARCHAR resource_id UUID old_values JSONB new_values JSONB ip_address VARCHAR user_agent TEXT created_at TIMESTAMP

Indexes actor_id resource_type created_at

13. SESSION ENTITY

Purpose Session login pengguna.

Fields id UUID PK user_id UUID FK token VARCHAR expires_at TIMESTAMP created_at TIMESTAMP
Indexes user_id expires_at

14. ENUM DEFINITIONS

Role CITIZEN OFFICER SUPER_ADMIN

ReportStatus DRAFT SUBMITTED ASSIGNED IN_PROGRESS RESOLVED CLOSED REJECTED

Priority LOW MEDIUM HIGH CRITICAL

ReportCategory INFRASTRUCTURE HEALTH EDUCATION SANITATION TRANSPORTATION OTHER

NotificationType INFO SUCCESS WARNING ERROR

15. DATABASE CONSTRAINTS

Unique users.email reports.report_number

Foreign Key reports.citizen_id reports.agency_id reports.location_id comments.user_id attachments.report_id
notifications.user_id

16. SOFT DELETE STRATEGY

Entities Users Reports Agencies

Fields deleted_at

Purpose Data recovery Audit compliance Historical analytics

17. ANALYTICS READY TABLES

Reports ReportHistory AuditLogs Notifications

These tables support: Resolution Rate Response Time Officer Performance Agency Performance Monthly Trends
Category Distribution

18. SCALING CONSIDERATIONS

Current Target 10,000 Users 100,000 Reports

Future Target 1,000,000+ Reports

Optimizations Indexes Query Optimization Read Replicas Redis Caching Partitioning

19. BACKUP STRATEGY

Daily Backup Weekly Snapshot Monthly Archive Retention 30 Days

20. SECURITY CONSIDERATIONS

Passwords Argon2

Sensitive Logs Encrypted

Audit Logs Immutable

Session Expiration Enabled